

Supervised Machine Learning Algorithms: Classification and Comparison

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1.ABSTRACT

Supervised Machine Learning (SML) is the search for algorithms that reason from externally supplied instances to produce general hypotheses, which then make predictions about future instances. Supervised classification is one of the tasks most frequently carried out by the intelligent systems. This paper describes various Supervised Machine Learning (ML) classification techniques, compares various supervised learning algorithms as well as determines the most efficient classification algorithm based on the data set, the number of instances and variables (features). Seven different machine learning algorithms were considered: Decision Table, Random Forest (RF), Naïve Bayes (NB), Support Vector Machine (SVM), Neural Networks (Perceptron), JRip and Decision Tree (J48) using Waikato Environment for Knowledge Analysis (WEKA) machine learning tool.

2. INTRODUCTION

Machine learning is one of the fastest growing areas of computer science, with far-reaching applications. It refers to the automated detection of meaningful patterns in data. Machine learning tools are concerned with endowing programs with the ability to learn and adapt.

Machine Learning has become one of the mainstays of Information Technology and with that, a rather central, albeit usually hidden, part of our life. With the ever-increasing amounts of data becoming available there is a good reason to believe that smart data analysis will become even more pervasive as a necessary ingredient for technological progress. There are several applications for Machine Learning (ML), the most significant of which is data mining. People are often prone to making mistakes during analyses or, possibly, when trying to establish relationships between multiple features.

Data Mining and Machine Learning are Siamese twins from which several insights can be derived through proper learning algorithms. There has been tremendous progress in data mining and machine learning as a result of evolution of smart and Nano technology which brought about curiosity in finding hidden patterns in data to derive value. The fusion of statistics, machine learning, information theory, and computing has created a solid science, with a firm mathematical base, and with very powerful tools. Machine learning algorithms are organized into a taxonomy based on the desired outcome of the algorithm. Supervised learning generates a function that maps inputs to desired outputs.

ML is perfectly intended for accomplishing the accessibility hidden within Big Data. ML hand over's on the guarantee of extracting importance from big and distinct data sources through outlying less dependence scheduled on individual track as it is data determined and spurts at machine scale. Machine learning is fine suitable towards the intricacy of handling through dissimilar data origin and the vast range of variables as well as amount of data concerned where ML prospers on increasing datasets. The extra data supply into a ML structure, the more it be able to be trained and concern the consequences to superior value of insights. At the liberty from the confines of individual level thought and study, ML is clever to find out and show the patterns hidden in the data.

One standard formulation of the supervised learning task is the classification problem: The learner is required to learn (to approximate the behavior of) a function which maps a vector into one of several classes by looking at several input output examples of the function. Inductive machine learning is the process of learning a set of rules from instances (examples in a training set), or more generally speaking, creating a classifier that can be used to generalize from new instances. The process of applying supervised ML to a real-world problem is described in Figure 1.



Figure 1: The Processes of Supervised Machine Learning

3. LITERATURE REVIEW

A. Classification of Supervised Learning Algorithms According to, the supervised machine learning algorithms which deals more with classification includes the following. Linear Classifiers, Logistic Regression, Naïve Bayes Classifier, Perceptron, Support Vector Machine; Quadratic Classifiers, K-Means Clustering, Boosting, Decision Tree, Random Forest (RF); Neural networks, Bayesian Networks and so on.

1) Linear Classifiers:

Linear models for classification separate input vectors into classes using linear (hyperplane) decision boundaries. The goal of classification in linear classifiers in machine learning, is to group items that have similar feature values, into groups. stated that a linear classifier achieves this goal by making a classification decision based on the value of the linear combination of the features. A linear classifier is often used in situations where the speed of classification is an issue, since it is rated the fastest classifier. Also, linear classifiers often work very well when the number of dimensions is large, as in document classification, where each element is typically the number of counts of a word in a document.

2) Logistic regression:

This is a classification function that uses class for building and uses a single multinomial logistic regression model with a single estimator. Logistic regression usually states where the boundary between the classes exists, also states the class probabilities depend on distance from the boundary, in a specific approach. This moves towards the extremes (0 and 1) more rapidly when data set is larger. These statements about probabilities which make logistic regression more than just a classifier. It makes stronger, more detailed predictions, and can be fit in a different way; but those strong predictions could be wrong.

3) Naive Bayesian (NB) Networks:

These are very simple Bayesian networks which are composed of directed acyclic graphs with only one parent (representing the unobserved node) and several children (corresponding to observed nodes) with a strong assumption of independence among child nodes in the context of their parent. Thus, the independence model (Naive Bayes) is based on estimating. Bayes classifiers are usually less accurate than other more sophisticated learning algorithms (such as ANNs). However, performed a large-scale comparison of the naive Bayes classifier with state-of-the-art algorithms for decision tree induction, instance-based learning, and rule induction on standard benchmark datasets, and found it to be sometimes superior to the other learning schemes, even on datasets with substantial feature dependencies

4) Multi-layer Perceptron:

This is a classifier in which the weights of the network are found by solving a quadratic programming problem with linear constraints, rather than by solving a nonconvex, unconstrained minimization problem as in standard neural network training. Other well-known

algorithms are based on the notion of perceptron. Perceptron algorithm is used for learning from a batch of training instances by running the algorithm repeatedly through the training set until it finds a prediction vector which is correct on all of the training set. This prediction rule is then used for predicting the labels on the test set.

5) Support Vector Machines (SVMs):

These are the most recent supervised machine learning technique. Support Vector Machine (SVM) models are closely related to classical multilayer perceptron neural networks. SVMs revolve around the notion of a —margin—either side of a hyperplane that separates two data classes. Maximizing the margin and thereby creating the largest possible distance between the separating hyperplane and the instances on either side of it has been proven to reduce an upper bound on the expected generalisation error.

6) K-means:

Kmeans is one of the simplest unsupervised learning algorithms that solve the well-known clustering problem. The procedure follows a simple and easy way to classify a given data set through a certain number of clusters fixed a priori. K-Means algorithm is employed when labeled data is not available. General method of converting rough rules of thumb into highly accurate prediction rule. Given —weak— learning algorithm that can consistently find classifiers at least slightly better than random, say, accuracy $\geq 55\%$, with sufficient data, a boosting algorithm can provably construct single classifier with very high accuracy, say, 99%.

7) Decision Trees:

Decision Trees (DT) are trees that classify instances by sorting them based on feature values. Each node in a decision tree represents a feature in an instance to be classified, and each branch represents a value that the node can assume. Instances are classified starting at the root node and sorted based on their feature values. Decision tree learning, used in data mining and machine learning, uses a decision tree as a predictive model which maps observations about an item to conclusions about the item's target value. More descriptive names for such tree models are classification trees or regression trees.

8) Neural Networks:

Neural Networks (NN) that can actually perform a number of regression and/or classification tasks at once, although commonly each network performs only one. In the vast

majority of cases, therefore, the network will have a single output variable, although in the case of many-state classification problems, this may correspond to a number of output units (the post-processing stage takes care of the mapping from output units to output variables). Artificial Neural Network (ANN) depends upon three fundamental aspects, input and activation functions of the unit, network architecture and the weight of each input connection. Given that the first two aspects are fixed, the behavior of the ANN is defined by the current values of the weights. The weights of the net to be trained are initially set to random values, and then instances of the training set are repeatedly exposed to the net.

B. Features of Machine Learning Algorithms

Supervised machine learning techniques are applicable in numerous domains. A number of Machine Learning (ML) application-oriented papers can be found in. Generally, SVMs and neural networks tend to perform much better when dealing with multidimensions and continuous features. On the other hand, logic-based systems tend to perform better when dealing with discrete/categorical features. For neural network models and SVMs, a large sample size is required in order to achieve its maximum prediction accuracy whereas NB may need a relatively small dataset. There is general agreement that k-NN is very sensitive to irrelevant features: this characteristic can be explained by the way the algorithm works. Moreover, the presence of irrelevant features can make neural network training very inefficient, even impractical. Most decision tree algorithms cannot perform well with problems that require diagonal partitioning. The division of the instance space is orthogonal to the axis of one variable and parallel to all other axes. Therefore, the resulting regions after partitioning are all hyperrectangles. The ANNs and the SVMs perform well when multi-collinearity is present and a nonlinear relationship exists between the input and output features. Finally, Decision Trees and NB generally have different operational profiles, when one is very accurate the other is not and vice versa. On the contrary, decision trees and rule classifiers have a similar operational profile. SVM and ANN have also a similar operational profile. No single learning algorithm can uniformly outperform other algorithms over all datasets.

Table 1: Comparing learning algorithms (**** stars represent the best and * star the worst performance)

	Decision Trees	Neural Networks	Naïve Bayes	kNN	SVM	Rule-learners
Accuracy in general	**	***	*	**	****	**
Speed of learning with respect to number of attributes and the number of instances	***	*	****	****	*	**
Speed of classification	****	****	****	*	****	****
Tolerance to missing values	***	*	****	*	**	**
Tolerance to irrelevant attributes	***	*	**	**	****	**
Tolerance to redundant attributes	**	**	*	**	***	**
Tolerance to highly interdependent attributes (e.g. parity problems)	**	***	*	*	***	**
Dealing with discrete/binary/continuous attributes	****	*** (not discrete)	*** (not continuous)	*** (not directly discrete)	** (not discrete)	*** (not directly continuous)
Tolerance to noise	**	**	***	*	**	*
Dealing with danger of overfitting	**	*	***	***	**	**
Attempts for incremental learning	**	***	****	****	**	*
Explanation ability/transparency of knowledge/classifications	****	*	****	**	*	****
Model parameter handling	***	*	****	***	*	***

4. RESEARCH METHODOLOGY

The research data was obtained from National Institute of Diabetes and Digestive and Kidney Diseases which was made available online at University of California, Irvine website: <https://archive.ics.uci.edu/ml/machine-learning-databases/pima-indians-diabetes/> (2017). This data was chosen because of its accuracy and has also been anonymized (de-identified), therefore confidentiality is ensured. The number of Attributes is 8 with one class making it 9. All attributes are numeric-valued as follows:

1. Number of times pregnant
2. Plasma glucose concentration a 2 hours in an oral glucose tolerance test
3. Diastolic blood pressure (mm Hg)
4. Triceps skin fold thickness (mm)
5. 2-Hour serum insulin (mu U/ml)
6. Body mass index (weight in kg/(height in m)²)
7. Diabetes pedigree function
8. Age (years)
9. Class variable (0 or 1)

5. RESULTS AND DISCUSSION

A. Results

WEKA was used in the classification and comparison of the various machine learning algorithms. Table 3 shows the results with 9 attributes as well as parameters considered. MAE (Mean Absolute Error) is a measure of how close forecast or predictions are to the eventual outcome. Kappa Statistic is a metric that compares an observed accuracy with an expected accuracy (Random Chance) Time is the TIME taking to build the model. MAE (Mean Absolute Error) is a measure of how close forecast or predictions are to the eventual outcome.

Algorithm	Time (Sec)	Correctly Classified (%)	Incorrectly Classified (%)	Test Mode	Attributes	No of instances	Kappa statistic	MAE	Precision of YES	Precision of NO	Classification
Decision Table	0.23	72.3958	27.6042	10-fold cross-validation	9	768	0.3752	0.341	0.619	0.771	Rules
Random Forest	0.55	74.7396	25.2604	10-fold cross-validation	9	768	0.4313	0.3105	0.653	0.791	Trees
Naive Bayes	0.03	76.3021	23.6979	10-fold cross-validation	9	768	0.4664	0.2841	0.678	0.802	Bayes
SVM	0.09	77.3438	22.6563	10-fold cross-validation	9	768	0.4682	0.2266	0.740	0.785	Functions
Neural Networks	0.81	75.1302	24.8698	10-fold cross-validation	9	768	0.4445	0.2938	0.653	0.799	Functions
JRip	0.19	74.4792	25.5208	10-fold cross-validation	9	768	0.4171	0.3461	0.659	0.780	Rules
Decision Tree (J48)	0.14	73.8281	26.1719	10-fold cross-validation	9	768	0.4164	0.3158	0.632	0.790	Tree

B. Discussion

It was observed that all the algorithms have higher Kappa statistic compared to MAE (Mean Absolute Error). Also, correctly classified instances are higher than incorrectly classified instances. This is an indication that with higher data sets, the predictive analysis is more reliable. SVM and NB require large sample size in order to achieve maximum prediction accuracy as shown in the table 3, while Decision Tree and Decision Table have the least precision. instances and 6 attributes. The Kappa statistics for Neural Networks, JRip and J48 are lower compared to MAE and this does not portray precision and accuracy. This shows that with smaller datasets Neural Networks, JRip and J48 shows drastic reduction in the percentage of correctly classified instances in comparison to incorrectly classified instances. However, with smaller data set SVM and RF shows high accuracy and precision. Whereas Decision Table

built the model with more time compare to JRip and Decision Tree. Therefore, less time does not guarantee accuracy. If Kappa Statistic is less than Mean Absolute Error (MAE), the algorithm will not show precision and accuracy. It follows that, the algorithm which such characteristics cannot be used for that data set as it will not show precision and accuracy.

6. CONCLUSION

ML classification requires thorough fine tuning of the parameters and at the same time sizeable number of instances for the data set. It is not a matter of time to build the model for the algorithm only but precision and correct classification. Therefore, the best learning algorithm for a particular data set, does not guarantee the precision and accuracy for another set of data whose attributes are logically different from the other. However, the key question when dealing with ML classification is not whether a learning algorithm is superior to others, but under which conditions a particular method can significantly outperform others on a given application problem. Meta-learning is moving in this direction, trying to find functions that map datasets to algorithm performance.

To this end, meta-learning uses a set of attributes, called meta-attributes, to represent the characteristics of learning tasks, and searches for the correlations between these attributes and the performance of learning algorithms. Some characteristics of learning tasks are: the number of instances, the proportion of categorical attributes, the proportion of missing values, the entropy of classes, etc. provided an extensive list of information and statistical measures for a dataset. After a better understanding of the strengths and limitations of each method, the possibility of integrating two or more algorithms together to solve a problem should be investigated. The objective is to utilize the strengths of one method to complement the weaknesses of another.

If we are only interested in the best possible classification accuracy, it might be difficult or impossible to find a single classifier that performs as well as a good ensemble of classifiers', NB and RF machine learning algorithms can deliver high precision and accuracy regardless of the number of attributes and data instances. This research shows that time to build a model is one factor on one hand; and precision with kappa statistic while MAE is another factor on the other hand. Therefore, ML algorithms requires precision, accuracy and minimum error to have supervised predictive machine learning.

7. REFERENCES

[1]

Alex S.& Vishwanathan, S.V.N. (2008). Introduction to Machine Learning. Published by the press syndicate of the University of Cambridge, Cambridge, United Kingdom. Copyright © Cambridge University Press 2008. ISBN: 0- 521-82583-0. Available at KTH website:
<https://www.kth.se/social/upload/53a14887f276540ebc81aec3/online.pdf> Retrieved from website: <http://alex.smola.org/drafts/thebook.pdf>

[2]

Bishop, C. M. (1995). Neural Networks for Pattern Recognition. Clarendon Press, Oxford, England. 1995. Oxford University Press, Inc. New York, NY, USA ©1995 ISBN:0198538642 Available at:
http://cs.du.edu/~mitchell/mario_books/Neural_Networks_for_Pattern_Recognition_-_Christopher_Bishop.pdf

[3]

Brazdil P., Soares C. & da Costa, J. (2003). Ranking Learning Algorithms: Using IBL and Meta-Learning on Accuracy and Time Results. Machine Learning Volume 50, Issue 3, 2003. Copyright © Kluwer Academic Publishers. Manufactured in The Netherlands, doi:10.1023/A:1021713901879pp. 251–277. Available at Springer website: <https://link.springer.com/content/pdf/10.1023%2FA%3A1021713901879.pdf>

[4]

Cheng, J., Greiner, R., Kelly, J., Bell, D. & Liu, W. (2002). Learning Bayesian networks from data: An information theory based approach. Artificial Intelligence Volume 137. Copyright © 2002. Published by Elsevier Science B.V. All rights reserved pp. 43 – 90. Available at science Direct:
<http://www.sciencedirect.com/science/article/pii/S0004370202001911>

[5]

Domingos, P. & Pazzani, M. (1997). On the optimality of the simple Bayesian classifier under zero-one loss. Machine Learning Volume 29, pp. 103–130 Copyright © 1997 Kluwer Academic Publishers. Manufactured in The Netherlands. Available at University of Trento website:
<http://disi.unitn.it/~p2p/RelatedWork/Matching/domingos97optimality.pdf>

- [6] Elder, J. (n.d). Introduction to Machine Learning and Pattern Recognition. Available at LASSONDE University EECS Department York website:
http://www.eecs.yorku.ca/course_archive/201112/F/44045327/lectures/01%20Introduction.pdf
- [7] Newsom, I. (2015). Data Analysis II: Logistic Regression. Available at:
http://web.pdx.edu/~newsomj/da2/ho_logistic.pdf
- [8] Hormozi, H., Hormozi, E. & Nohooji, H. R. (2012). The Classification of the Applicable Machine Learning Methods in Robot Manipulators. International Journal of Machine Learning and Computing (IJMLC), Vol. 2, No. 5, 2012 doi: 10.7763/IJMLC. 2012.V2.189pp. 560 – 563. Available at IJMLC website:
<http://www.ijmlc.org/papers/189-C00244-001.pdf>
- [9] Kotsiantis, S. B. (2007). Supervised Machine Learning: A Review of Classification Techniques. Informatica 31 (2007). Pp. 249 – 268. Retrieved from IJS website:
<http://wen.ijs.si/ojs2.4.3/index.php/informatica/article/download/148/140>
- [10] Lemnaru C. (2012). Strategies for dealing with Real World Classification Problems, (Unpublished PhD thesis) Faculty of Computer Science and Automation, Universitatea Technica, Din Cluj-Napoca. Available at website:
<http://users.utcluj.ro/~cameliav/documents/TezaFinalLemna ru.pdf>
- [11] Logistic Regression pp. 223 – 237. Available at:
<https://www.stat.cmu.edu/~cshalizi/uADA/12/lectures/ch12 .pdf>
- [12] Rob Schapire (n.d) Machine Learning Algorithms for Classification.
- [13] Rob Schapire (n.d) Machine Learning Algorithms for Classification.
- [14] ssSetiono R. and Loew, W. K. (2000), FERNN: An algorithm for fast extraction of rules from neural networks, Applied Intelligence